

EDUCATION

Carleton University

Sep 2023 – Jan 2025

Master of Engineering: Electrical and Computer Engineering (Data Science) GPA - 10.33/12.00

SKILLS

Languages	Python, SQL, Java, JavaScript, C/C++, MATLAB, HTML/CSS
Frameworks & Libraries	TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, OpenCV, SpaCy, FastAPI, Mediapipe
Tools	Git, Linux, Docker, GCP, Looker Studio, Tableau, Power BI, Postman
Web & API	Node.js, Express.js, REST APIs, Google Calendar API
Platform	Google Cloud Platform (BigQuery, Compute Engine), Streamlit, MongoDB

EXPERIENCE

Computer Science Student

Jul 2024 — Current

Ottawa Cognitive Centre for Therapy

- Implemented RESTful API integration between the organization's scheduling software and CRM/EHR systems, enhancing cross-platform data flow and improving user interaction efficiency by 50%.
- Designed and managed JSON-based data structures for client scheduling workflows, streamlining data exchange between frontend and backend services, resulting in a 30% increase in system responsiveness.

Research Intern

Dec 2022 — May 2023

Space Application Centre (ISRO)

- Utilized Google Earth Engine (GEE) and Python to analyze MODIS, CHIRPS, ERA5/CFSR, and TRMM satellite datasets, generating detailed visualizations of hydrological indicators (evaporation, soil moisture, precipitation) to uncover trends across major Indian river basins.
- Implemented and compared machine learning models, including XGBoost, LSTM, and Random Forest to predict river discharge and hydraulic parameters, improving the accuracy of groundwater recharge estimation by 25%.

Machine Learning Intern

Dec 2022 — Feb 2023

Jupiter AI Labs Pvt. Ltd.

part-time

- Designed a series of practical case studies covering SQL, Excel-based analytics, exploratory data analysis, hypothesis testing, probability distributions, and data visualization using Python and Tableau, enhancing team-wide hands-on proficiency in applied data science.
- Acted as an AI prompt engineer, creating and refining advanced Python code through AI tools for internal and client projects, improving code efficiency by 40% and achieving a 95% client satisfaction rate.

PROJECTS

Multi-label Dataset Prediction | Python, Machine Learning

- Implemented advanced data preprocessing and feature selection techniques on large datasets (25K rows, 1024 features), enhancing model performance for a multi-label classification task.
- Applied logistic regression with multi-output classifier and meta-learning strategies, achieving a 75.00% F1-score and securing 5th place in a project competition.

Digital Trainer for Yoga Pose Estimation | Python, Deep Learning, Computer Visio, Mediapipe

- Developed a solution for detecting and predicting yoga asanas using various pose detection methodologies.
- Engineered an AI-driven landmark detection system using MediaPipe; enhanced keypoint identification accuracy by 25%, resulting in a more responsive and user-friendly customer interaction experience within the application.

Tone Topic: Topic Modeling and Labeling in the Streamlit Sea | Python, Machine Learning, Streamlit

- Engineered a robust Streamlit-based topic modeling and tagging tool, leveraging LDA and Heapq algorithms; streamlined market research efforts by processing over 1,000 textual documents in under five minutes.

Student Grievance Redressal Portal | NodeJS, ExpressJS, MongoDB, ReactJS

- Designed and developed a web portal that streamlined the student grievance process, resulting in a significant reduction in resolution time for complaints, and enhancing overall student satisfaction with administrative services.

PUBLICATIONS

Yoga Pose Estimation using Angle Heuristic Approach Published in 2022 4th International Conference on Inventive Research in Computing Applications (ICIRCA)